

Lab 3: Component Modelling & Architectural Pattern Selection

Written Justification — Peer Skill Exchange & Mentorship Network

Architecture Selection

We chose Layered Architecture for the Peer Skill Exchange & Mentorship Network. The system is divided into Presentation, Business, and Data / Integration layers so that user interaction, application logic, and persistent data are separated.

Architectural Choice

The Presentation Layer contains the Member Interface Component for the web/mobile interface. The Business Layer contains the Skill & Mentor Service, Session Management Component, Time Credit & Verification Component, and External Calendar Service. The Data / Integration Layer contains the Database Component. Components communicate through defined provided and required interfaces, while dashed «use» dependencies show service usage.

Two Reasons for the Choice

1. Clear separation of responsibilities: mentor discovery, session scheduling, credit verification, calendar interaction, and data storage are handled by focused components, making the system easier to understand, test, and modify.
2. Controlled component interaction: the member interface communicates with business services through interfaces instead of directly accessing stored data. This reduces coupling and allows individual components to be changed without redesigning the whole system.

Security Advantage

The Member Interface does not directly access the database or credit records. Requests pass through business components, where authentication, authorization, session validation, and credit-verification checks can be applied before sensitive operations are performed. This keeps access to member and credit information controlled through the application layer.

Performance Benefit

Frequently accessed skill and mentor information can be handled by the Skill & Mentor Service, while database access is kept behind the database interface. Session scheduling and calendar export are isolated behind interfaces, so external calendar operations do not have to be tightly coupled to the member-facing interface.

Diagram Components and Interfaces

The diagram contains the Member Interface Component, Skill & Mentor Service, Session Management Component, Time Credit & Verification Component, Database Component, and External Calendar Service. The main interfaces shown are Skill Search API, Session API, Credit / Verify API, Database Query Interface, Member / Skill Data, Credit Data, and iCal Calendar Export. Solid connectors represent interface connections, and dashed «use» arrows represent component dependencies.

Trade-off

Layered architecture introduces some additional interaction between layers and makes independent scaling of individual components less direct. For this system, the separation and controlled interfaces provide a clear structure for development and maintenance, making the trade-off acceptable.